Brian Won

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Education

Class 2025 **B.Sc. Computer Science & Statistics — cGPA: 3.84/4.0**, *University of Toronto*, Toronto, ON, Honours.

- Teaching Assistant: **CSC369 Operating Systems** | Graduate Coursework: **CSC413 Neural Networks & Deep Learning**

Audit **Deep Learning Systems**, *Carnegie Mellon University*.

Technical Skills

Programming Python, C, C++, CUDA, Rust, Java, TypeScript, SQL

Frameworks PyTorch, TensorRT-LLM, vLLM, Triton, needle (custom autograd), DDP, FSDP, DeepSpeed ZeRO, RLHF

Infrastructure Linux, Docker, Kubernetes, Terraform, CI/CD, Airflow, FastAPI, Redis, PostgreSQL, Prometheus, Grafana, Nsight Systems

Professional Experience

2026 **Software Engineer, Pre-AI Training (Contractor)**, *LinkedIn*, Remote.

- Engineered structured evaluation harnesses and automated scoring rubrics for code generation tasks (Python, TypeScript, SQL); authored adversarial test cases targeting correctness, edge-case robustness, and efficiency—outputs used directly in LinkedIn's internal LLM training pipeline.
- Constructed instruction-following datasets by designing paired prompt-completion examples across software engineering domains; applied systematic code quality criteria to produce clean, high-signal training signal for fine-tuning.

2025 **Undergraduate Researcher**, *Ontario Teachers' Association*, Toronto, ON.

- Instrumented end-to-end **monitoring and logging pipelines** (Prometheus, Grafana) for a platform under 18k+ concurrent users; established latency baselines and automated regression detection—directly applicable to ML training pipeline observability.
- Built **Docker**-based CI/CD provisioning and health-check automation cutting deployment cycle time **40%**; applied reproducible environment configs across the fleet, mirroring ML experiment reproducibility practices.

2023 **Systems Engineer**, *RBC Capital Markets*, New York City.

- Profiled and optimized latency-sensitive data pipelines in trading risk/pricing infrastructure, achieving **70% end-to-end latency reduction**; reworked caching layers and fetch paths on P&L-sensitive flows.
- Produced **quantified benchmark analysis** that directly drove architectural decisions across a globally distributed team—the same data-backed decision-making expected in large-scale AI infrastructure roles.

Selected Projects

2024 **Deep Learning Network Engine**, *C++ · CUDA · Python · numpy*.

- **CNN + CUDA:** Implemented `flip`, `pad`, `dilate`, and full **Conv** (forward & backward) atop the CUDA backend; built **ResNet9** for CIFAR-10 classification.
- **Transformer:** Multi-head self-attention and cross-attention (prenorm) from scratch; full Transformer encoder/decoder stack validated on sequence modeling tasks.

2025 **Custom CUDA Kernel & GPU Profiling**, *CUDA C++ · Nsight Compute · Roofline Analysis*.

- Implemented tiled shared-memory matrix multiplication kernel in CUDA, reaching **14.2 TFLOP/s** (87% of cuBLAS peak) and **823 GB/s** effective memory bandwidth—a **3.1**× speedup over naive;

2025 **LLM Inference Optimization: Tensor Kernels**, *TensorRT-LLM · Triton · CUDA · PyTorch · vLLM*.

- Deployed Llama-2-7B through **TensorRT-LLM** with INT8 weight-only quantization and in-flight batching; benchmarked against vLLM across batch sizes 1–64, achieving **2.3**× throughput and **38% P99 latency reduction** at batch size 16; applied **FlashAttention-2** and **FP8** to attention layers, cutting per-token latency a further **27%** on A100.
- Authored a fused **softmax + dropout** kernel in **Triton**; measured **1.8**× speedup via Nsight Compute with memory bandwidth improving from 61% to 89% of peak—directly applicable to TensorRT operator fusion; profiled full inference graph via **Nsight Systems** to surface attention and layernorm as primary bottlenecks.